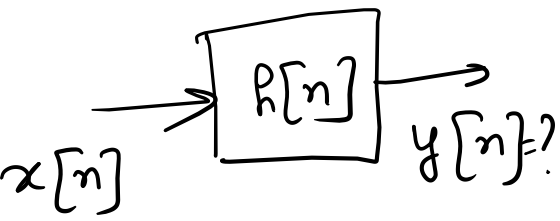
8|9|2025 Ex $i/p = x[n] = \begin{cases} 1 & 0 \le n < 4 \\ 0 & \text{otherwise} \end{cases}$

Impulse response of a D.T. LTI system $\Rightarrow h[n] = \begin{cases} \alpha^n & 0 \le n \le 6 \\ 0 & \text{otherwise} \end{cases}$

$\alpha > 1$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

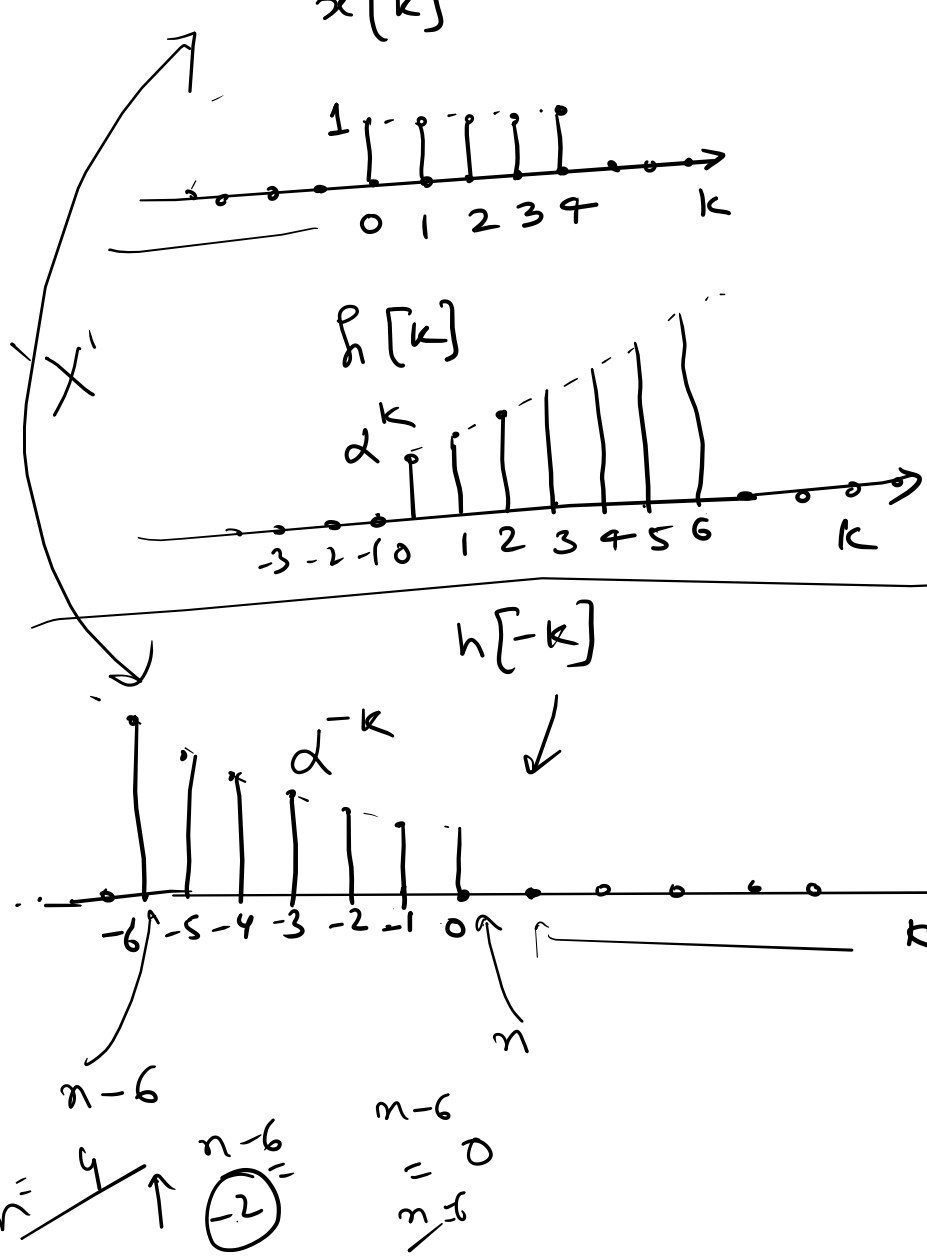


$x[n]$ → $x[k]$

$h[n]$ → $h[k]$ → $h[-k]$

$n = -ve$ ↓ shift $h[-k]$ towards left side by amount of n

if $n = +ve$ shift $h[-k]$ towards right side by the amount of n.



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] \Longleftarrow$$

$n = 0$

$$y[0] = \sum_{k=-\infty}^{\infty} x[k]\, h[-k]$$

$$= \ldots + x[-6]h[6] + x[-5]h[5] + \ldots$$
$$+ x[-1]h[1] + x[0]h[0]$$
$$+ x[1]h[-1] + \ldots\ldots$$
$$= x[0]h[0] = 1$$

$n > 0$ — shift $h[-k]$ to the right by the same amount

① when $n = 0$ to 4, my overlap is from $k = 0$ to 4 (0 to n)

② when $n = 5$ to 6, overlap is from 0 to 4

③ when $n = 7$ to 10, overlap is from $n-6$ to 4

④ $n \geq 11$, $y[n] = 0$

& ⑤ $n < 0$, $y[n] = 0$

Case-1

$0 \le n \le 4$



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=0}^{n} \alpha^{n-k}$$

$n-k=r$

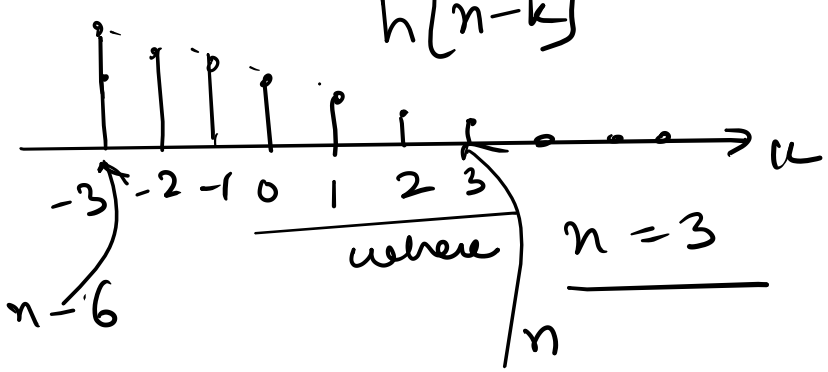
if $k=0$

$\Rightarrow r = n$

if $k=n$

$r=0$

$$= \sum_{r=0}^{n} \alpha^{r} = \frac{\alpha^{n+1}-1}{\alpha-1}$$



where $n=3$

$$y[0] = 1$$

$$y[1] = 1+\alpha$$

$$y[2] = 1+\alpha+\alpha^2$$

$$y[3] = 1+\alpha+\alpha^2+\alpha^3$$

$$y[4] = 1+\alpha+\alpha^2+\alpha^3+\alpha^4$$

$n = 5$

Case-2

$5 \le n \le 6$ $y[n]$

when $n=5$ Trailing edge of $h[n-k]$ is at $5-6=-1$

when $n=6$ " " $6-6=0$

Case-2
$n=5$

$h[n-k]$

$n-6$ to n

$n=5$: -1 to 5

$n=6$: 0 to 6

$n=7$: 1 to 7

$n=8$: 2 to 8

② Case-2 $5 \le n \le 6$

$$y[n] = \sum_{k=0}^{4} \alpha^{n-k}$$

③ Case-3: $6 \le n \le 10$

$$y[n] = \sum_{k=n-6}^{4} \alpha^{n-k}$$

④ Case-4: $n \ge 11$

$n = 11$, $n-6$ is at $11-6 = 5$ ⇒ tail-ip edge of $h[n-k]$ is at 5, way past non-zero values of $x[k]$

$y[n] = 0$

⑤ $n < 0$

leading edge of $h[n-k]$ is at n. It is outside non-zero values of $x[k]$

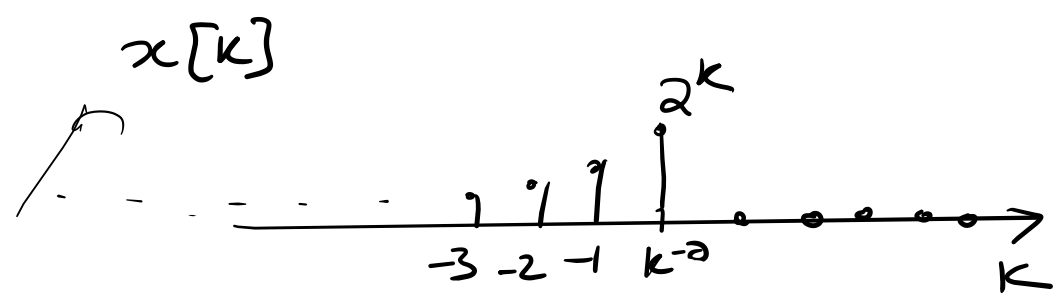
hence, $y[n] = 0$

$$y[n] = \begin{cases} 0 & n < 0 \ \& \ n \ge 11 \\ \sum_{k=0}^{n} \alpha^{n-k} & 0 \le n \le 4 \\ \sum_{k=0}^{4} \alpha^{n-k} & 5 \le n \le 6 \\ \sum_{k=n-6}^{4} \alpha^{n-k} & 7 \le n \le 10 \end{cases}$$

Ex

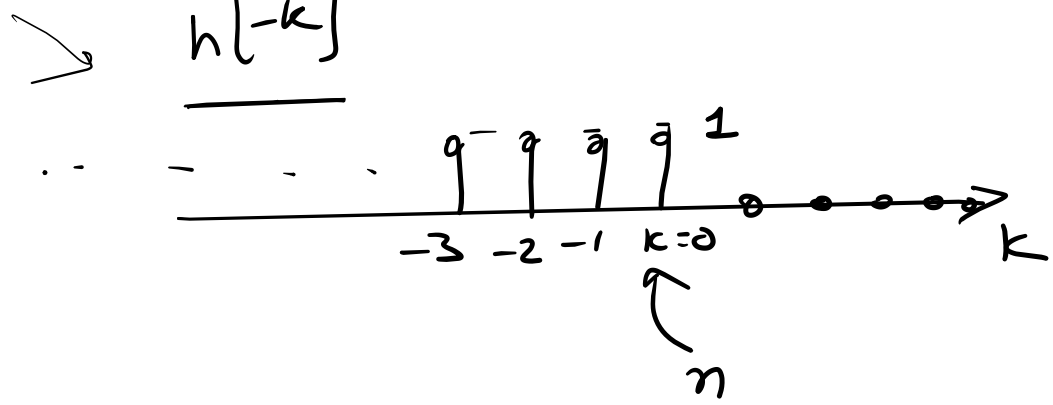
$$x[n] = 2^n u[-n]$$

$$h[n] = u[n]$$



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$n \geq 0$



Case-1
non-zero

overlap between $x[k]$ & $h[n-k]$ is from $k = -\infty$ to $k = 0$

$$y[n] = \sum_{k=-\infty}^{0} 2^k = \sum_{k=0}^{\infty} 2^{-k} = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{1-\frac{1}{2}} = 2$$

Case-2

$n < 0$

Let $n = -1$

$$y[n] = \sum_{k=-\infty}^{-1} 2^k = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k = 2 - 1 = 1$$

$n = -2$

$$y[n] = \sum_{k=-\infty}^{2} 2^k = \sum_{k=2}^{\infty} \left(\frac{1}{2}\right)^k$$

$\underline{n < 0}$

$$y[n] = \sum_{k=-n}^{\infty} \left(\frac{1}{2}\right)^k$$

$$y[n] = \begin{cases} 2 & ; \quad n \geq 0 \\ \sum_{k=-\infty}^{n} 2^k = \sum_{k=-n}^{\infty} \left(\frac{1}{2}\right)^k & ; \quad n < 0 \end{cases}$$

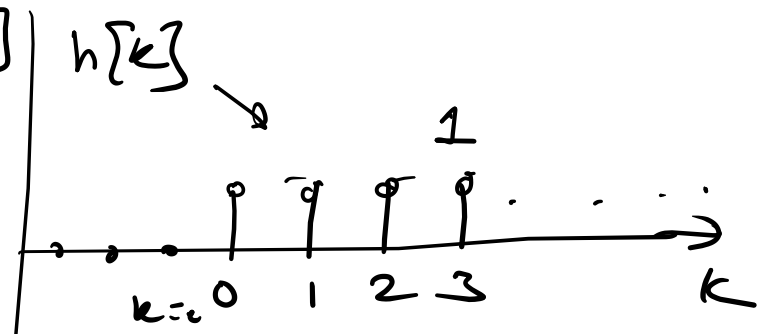
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = \sum_{r=-\infty}^{\infty} h[r]\, x[n-r] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

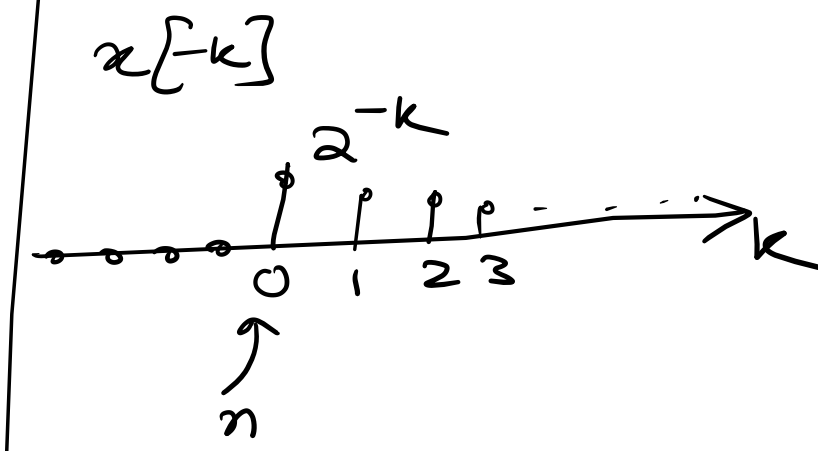
On R.H.S. Let $n - k = r$; when $k = -\infty$, $r = n + \infty = \infty$

$\boxed{n - r = k}$ when $k = +\infty$, $r = n - \infty = -\infty$

$x[n] = 2^n u[-n]$ & $h[n] = u[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$ Convolution Sum





Case 1 $n \geq 0$

$$y[n] = \sum_{k=n}^{\infty} 2^{n-k} = 2^n \sum_{k=n}^{\infty} 2^{-k}$$

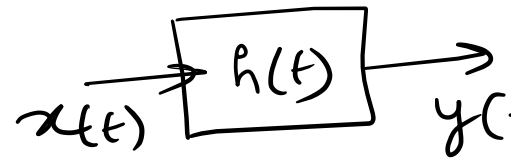
Case-2 $n < 0$

$$y[n] = \sum_{k=0}^{\infty} 2^{n-k} = 2^n \sum_{k=0}^{\infty} 2^{-k}$$

$x[n-k] = 2^{n-k} u[k+n]$

$x[-k+n]$

C.T. LTI



LTI

$h(t)$ denotes the impulse response of the system.

This implies that

if $x(t) = \delta(t)$

then $y(t) = h(t)$

$$y(t) = x(t) * h(t) = h(t) * x(t)$$

In general

$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

convolutional integral.

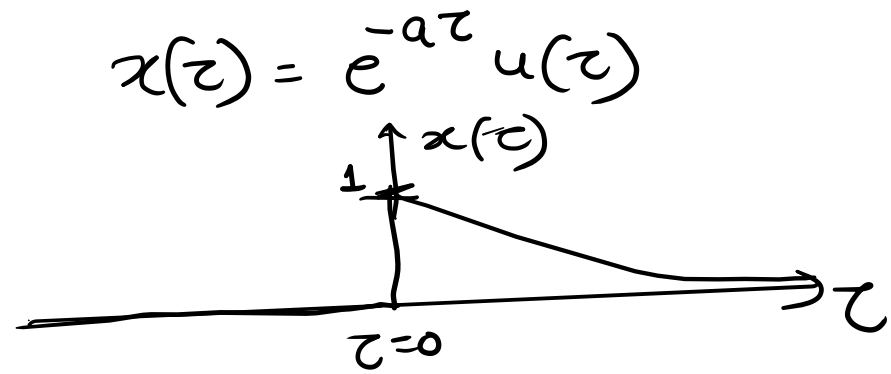
"$*$" — Linear & commutative operator

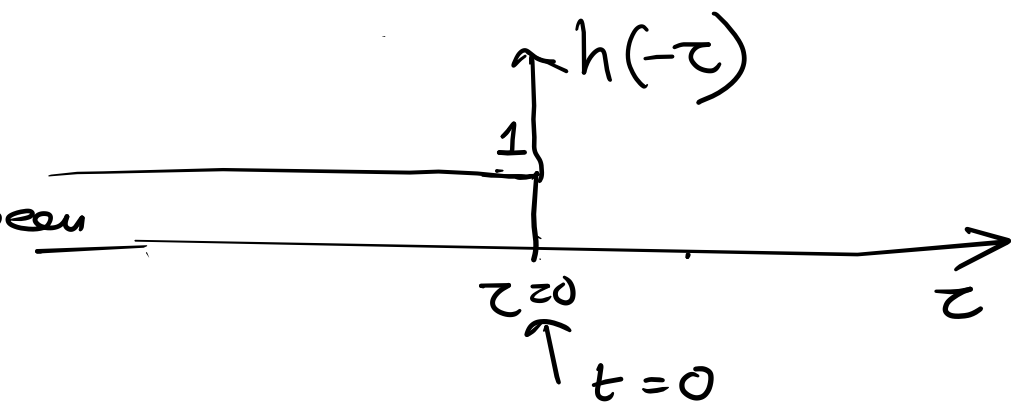
$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

Ex $x(t) = e^{-at} u(t)$; $a > 0$

$h(t) = u(t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$





① Case 1 $t < 0$

$y(t) = 0$ ∵ non-zero overlap between $x(\tau)$ & $h(t-\tau)$ is zero.

② Case 2 $t \geq 0$

$$y(t) = \int_0^t e^{-a\tau}\, d\tau = \frac{e^{-a\tau}}{-a}\Big|_0^t = \left(\frac{1 - e^{-at}}{a}\right)$$

Ex $x(t) = u(t)$ & $h(t) = e^{-at}u(t)$; $a > 0$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

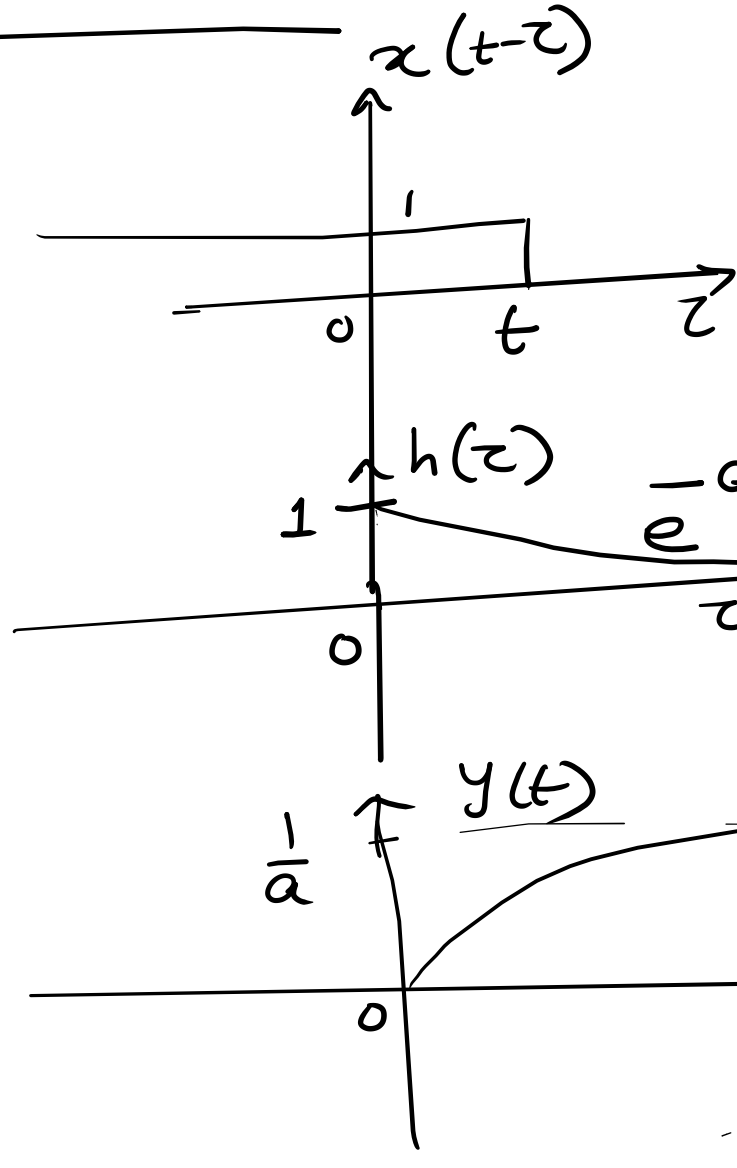
Case 1 $t < 0$

$$y(t) = 0$$

Case 2 $t \geq 0$

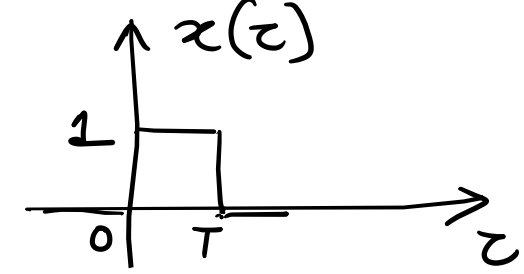
$$y(t) = \int_{0}^{t} e^{-a\tau}\, d\tau = \left(\frac{1 - e^{-at}}{a}\right)$$

$$y(t) = \begin{cases} 0 & ; \ t < 0 \\ \left(\dfrac{1 - e^{-at}}{a}\right) & ; \ t \geq 0 \end{cases}$$



Ex Let $x(t) = \begin{cases} 1 & 0 \le t \le T \\ 0 & \text{otherwise} \end{cases}$ & $h(t) = \begin{cases} t & 0 \le t \le 2T \\ 0 & \text{otherwise.} \end{cases}$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

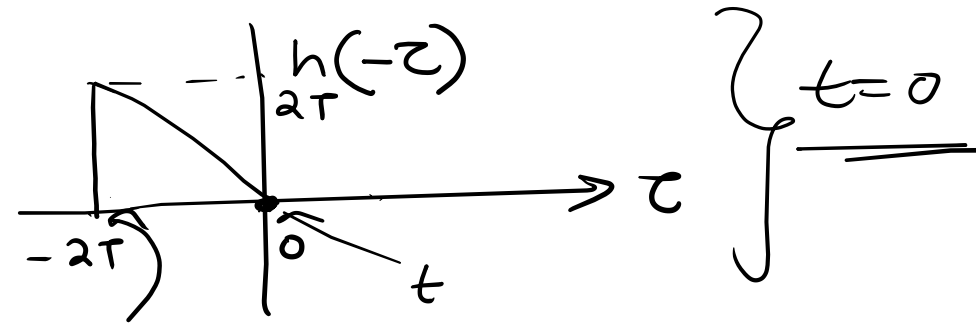


Cases

① Case-1 : $t < 0$

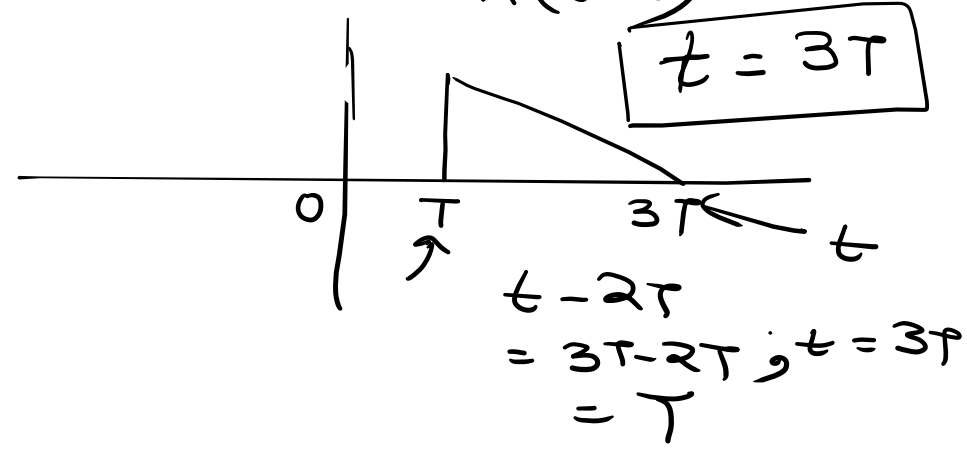
there is no non-zero overlap between $h(t-\tau)$ & $x(\tau)$. Hence,

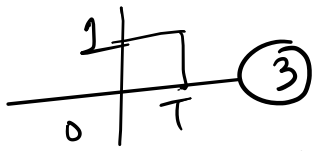
$y(t) = 0$



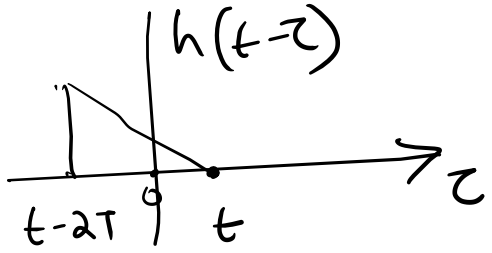
② Case-2 : $t > 3T$

$y(t) = 0$





③



Case-3 $\quad 0 \le t \le T$

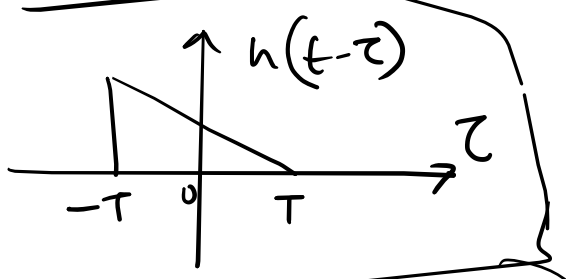
$$y(t) = \int_0^t x(\tau)\, h(t-\tau)\, d\tau = \int_0^t (t-\tau)\, d\tau = t\tau - \frac{\tau^2}{2}\Big|_0^t$$

$$= t^2 - \frac{t^2}{2} = \frac{t^2}{2}$$

④

Case-4 $\quad T \le t \le 2T$

t = T

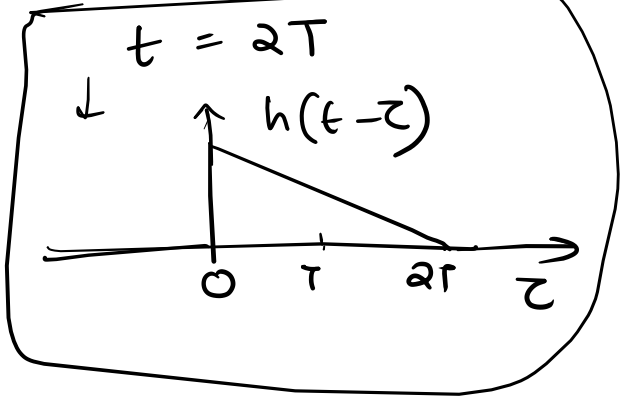


$$y(t) = \int_0^T x(\tau)\, h(t-\tau)\, d\tau = t\tau - \frac{\tau^2}{2}\Big|_0^T$$

$$= tT - \frac{T^2}{2}$$

Case 5 : $\quad 2T \le t \le 3T$

t = 2T

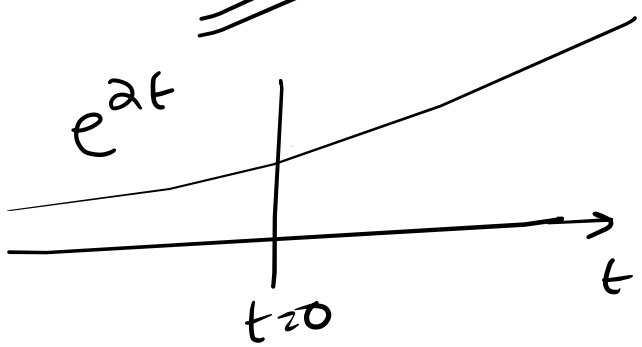


$$y(t) = \int_{t-2T}^{T} x(\tau)\, h(t-\tau)\, d\tau = t\tau - \frac{\tau^2}{2}\Big|_{t-2T}^{T}$$

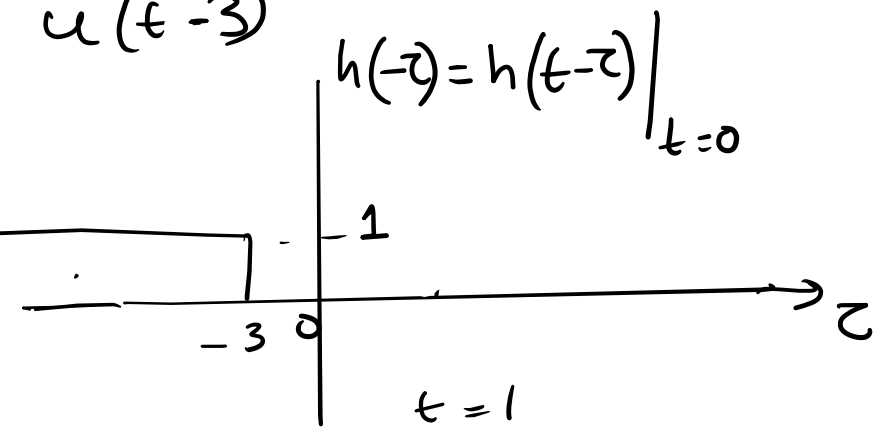
$$= \left(tT - \frac{T^2}{2}\right) - \left[t(t-2T) - \frac{(t-2T)^2}{2}\right]$$

Ex

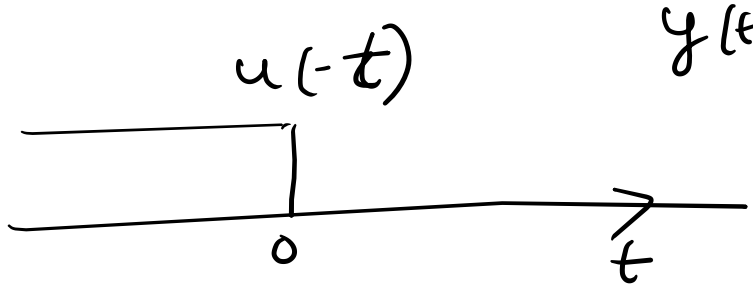
$x(t) = e^{2t}\,u(-t)$ & $h(t) = u(t-3)$



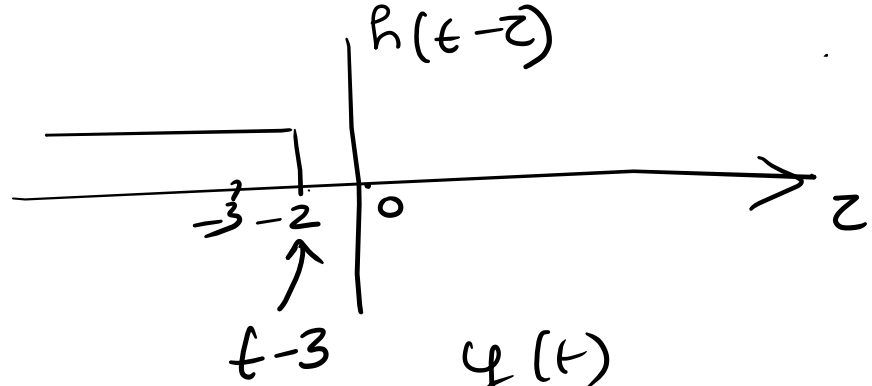
$$y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$



Case-1 : $t-3 < 0 \Rightarrow t < 3$

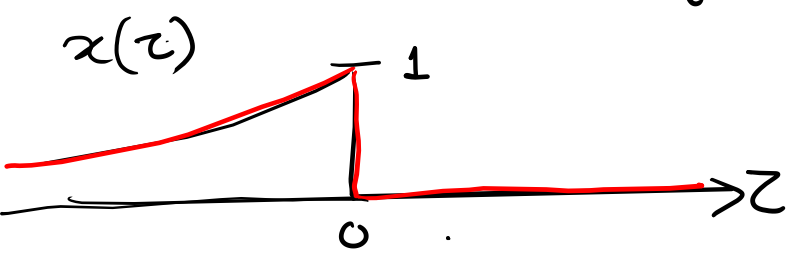


$$y(t) = \int_{-\infty}^{t-3} x(\tau)\,d\tau$$

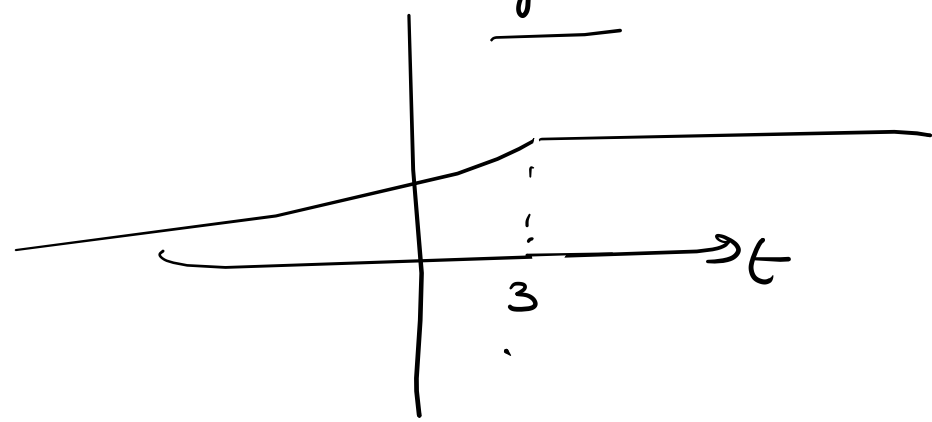


Case-2 : $t-3 \geq 0$

$t \geq 3$

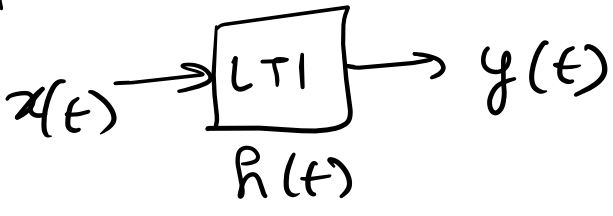


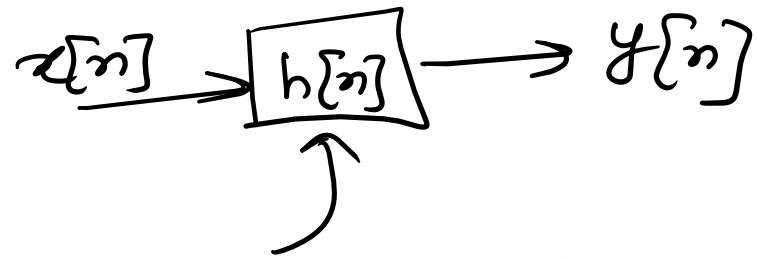
$$y(t) = \int_{-\infty}^{0} x(\tau)\,d\tau$$



Properties of LTI System —

① LTI systems are fully characterized by their impulse response.





uniquely characterized $h[n]$ ⟺ uniquely characterize the LTI system. $\neq h[n]$

a) $y[n] = x[n] + x[n-1]$

b) $y[n] = \max\{x[n], x[n-1]\}$

c) $y[n] = (x[n] + x[n-1])^2$

$\Rightarrow y[n] = \delta[n] + \delta[n-1]$ if $x[n] = \delta[n]$

① System is memoryless : $h(t) = k\delta(t)$

$y[n] = k\,x[n] \Rightarrow h[n] = k\,\delta[n]$

if $k = 1$, it is an identity system

② System is causal ⟶

$h(t) = 0, \; t < 0$

General case $\Rightarrow y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau = \int_{-\infty}^{\infty} h(\tau)\,x(t-\tau)\,d\tau$

$\tau \le t$

$= \int_{-\infty}^{t} x(\tau)\,h(t-\tau)\,d\tau$

$t - \tau = r, \quad r = \infty$

$\tau = -\infty, \quad r = 0$

$\tau = t, \quad t - r = \tau$

$= \int_{-\infty}^{\infty} x(t-r)\,h(r)\,dr = \int_{0}^{\infty} h(\tau)\,x(t-\tau)\,d\tau$